Science Connections: Evidence Control

I can connect a scientific question to the right idea and use evidence to explain my answer.

QUESTION
use evidence

EVIDENCE
use evidence

REASON
use evidence

[Open the original interactive lesson]

What success looks like

- I can identify whether a question is about forces, space or climate.
- I can select evidence that is relevant to the question.
- I can link an answer to evidence using a scientific reason.

Retrieve and reconnect

- State what friction does to movement.
- Explain one cause of day and night.
- Name the mechanism linking extra greenhouse gas to warming.
- Explain why repeated results are stronger than one result.

A rover travels less far on fabric than on a smooth surface. Which idea should control the answer?

Think first. Point, say, sign or write your choice. Give one reason.

A. Friction between surfaces

B. Earth's rotation

C. The enhanced greenhouse effect

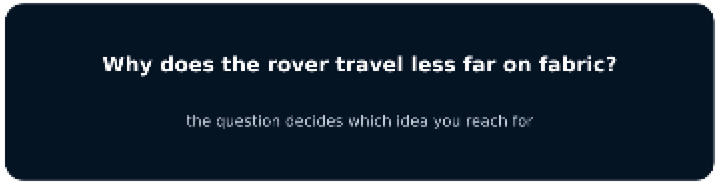
Look closely · what is the evidence?

QUESTION · what must I answer?

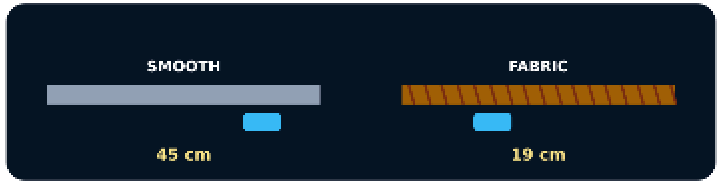
EVIDENCE · which result matters?

REASON · which idea explains it?

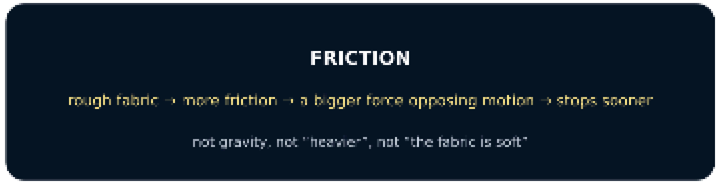
Look closely · what is the evidence?



The question



Rover distances on smooth and fabric



Friction reason

Words we will use

- evidence — what we observed
- connection — how ideas link
- cause — what makes a change happen

Watch the model

- Worked practice question: Why did the rover travel less far on fabric? Name the topic first: forces and friction. These practice data are not next week's investigation.

- Worked evidence: smooth 46, 44, 45 cm; fabric 18, 20, 19 cm. Every fabric result is lower.
- Reason: fabric creates more friction on the rover, opposing motion, so it travels a shorter distance.

Which answer best connects the rover question, evidence and reason?

A. The rover travelled less on fabric: 18–20 cm compared with 44–46 cm, because greater friction opposed its motion.

B. Earth rotates once in about 24 hours.

C. The fabric result looks best to me.

Commit to an answer. Show the evidence you used. Listen, then revise if needed.

Build the explanation

- Space answer: day and night happen because Earth rotates, moving a place into and out of sunlight.
- Climate answer: extra greenhouse gases absorb and re-emit more infrared energy, so the global average rises.
- For every topic, keep the same control route: question, relevant evidence, scientific reason, honest limit.

Find and repair the misconception

Every scientific question can use the same fact as evidence.

A. Keep it: any true fact answers any question.

B. Repair it: evidence must be relevant to the exact question.

C. Change it: the longest fact is always best.

Evidence Control Board

Place each evidence card under Forces, Space or Climate. Say the scientific link, not just the topic name.

- Read the question
- Choose the topic
- Select relevant evidence
- State the reason

Use the numbered cards in your pupil pack. Take turns to choose, justify and check. You may direct a partner to move a card.

Choose a group · defend your placement

- FORCES
- SPACE
- CLIMATE

Choose one frozen card and give a complete answer: claim, evidence, scientific reason and one limit.

Your independent mission

Answer the embedded questions: Forces — why did the rover travel less far on fabric? Space — why do day and night happen? Climate — why can extra greenhouse gas raise Earth's average temperature?

Record: At least one complete answer with relevant evidence, scientific reasoning and one honest limit.

Choose your support and challenge

- Supported: Choose one embedded question: rover and friction, day and night, or extra greenhouse gas and warming. Match one claim, evidence card and reason.
- Standard: Answer two of the three embedded questions from different topics and underline the evidence used.
- Stretch: Answer all three embedded topic questions, compare evidence quality and name one model or data limitation.

Show what you know

Use one result to answer a forces, space or climate question.

Point to, read or explain the evidence you made. Choose one next step after the adult has listened.